Where MUNI Loses Time

CP101 Final Project — Midterm Progress Report | November 5, 2026

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Github: https://github.com/SNORLAX223/CYPLAN101_Project_G16

Track: B

1. Research Question

Our project looks at how dependable MUNI travel times are in San Francisco. We want to know which parts of a bus or light-rail route take longer than the schedule says they should, and whether the delays change throughout the day.

We focus on weekdays in March 2025. We compare the scheduled travel time between two consecutive stops with the observed travel time for the same segment. We then group results into four periods: morning peak (6–9 a.m.), midday (9 a.m.–3 p.m.), afternoon peak (3–7 p.m.), and evening (7–10 p.m.).

Since our September proposal, we added a second part to the project: an equity question. We now compare MUNI delay patterns with nearby neighborhood characteristics, including household income, access to a car, and the share of commuters who use public transit. This helps us explore where reliability problems may matter most, while recognizing that neighborhood data cannot tell us exactly who was on a vehicle.

2. Data and Cleaning

MUNI service and travel times

Our main source is the 511 SF Bay historical transit feed for March 2025. The most important file is stop_observations.txt, which includes scheduled and observed times for MUNI stop visits. We also use stops.txt to locate stops on a map.

To clean the file, we first filtered to Muni only, removing around 55% of the rows. We then filtered down to only March 2025 weekdays, removing around 28.4% of the rows. Both of these steps were done to keep the data within the scope of the research question. Following that, we removed any duplicate stop visits (0.7%) to make sure no visits are double counted. The trip_start_time column was dropped as well, since there was no data in 100% of the rows. The time strings were then converted to seconds, and then we dropped rows that can't build segments. This removed around 2.7% of the data, which includes all the first stops of each trip, and any row missing a needed time. Looking at the segments, we saw that 2 rows had a mismatch from_stop_id and to_stop_id, meaning that there was a gap in the sequence. We decided to drop these 2 rows, so that every segment was truly two consecutive stops. After further inspection, we also decided to drop impossible values of negative observed travel time, and rows with a delay over 30 minutes, which covers less than 1% of the data. Dropping these rows allows us to look at more systemic issues, as delays this long are either human error or caused by outside circumstances. Our final step was removing any rows that fall outside of the 4 time periods, which accounts for about 13% of the data.

Neighborhood information

Our second source is the 2024 American Community Survey (ACS) five-year estimates for San Francisco census tracts. We use median household income, the percentage of households without a vehicle, and the percentage of workers who commute by public transit. The cleaning process was straightforward. The Census reports missing data as large negative values, so we converted these to

“not available” to not skew the median. We then created percentage columns to compare the different categories, and saw that some tracts have zero households, therefore having no percentages. 10 tracts were flagged having an income at the ACS cap of $250,001, which we converted to the true value of “250,001” or more. We then obtained tract boundaries separately using pygris. The geographic analysis matched 242 mapped tracts to ACS data and identified 3,234 MUNI stops used in our segment dataset.

A small number of stops do not have usable values for some Census measures. We keep these gaps visible rather than making up values. We will report the exact number of unmatched or missing records in the final cleaning summary.

3. Methods

We calculate the scheduled travel time for each stop-to-stop segment by subtracting the scheduled departure time at the first stop from the scheduled arrival time at the next stop. We calculate observed travel time in the same way using actual departure and arrival times. Our main measure is:

$$\text{Travel-time delay} = \text{observed segment time} - \text{scheduled segment time}$$

A positive value means the vehicle took longer than planned. A negative value means it was faster than planned. This measures time spent moving between stops, not the number of minutes a vehicle arrives late at the end of a trip.

Next, we group observations by route, direction, starting stop, ending stop, and time period. For each group, we calculate the median delay, average delay, number of observations, number of service days, and percentage of observations slower than scheduled. We only keep summary groups with at least 30 observations spread across at least five service days. This leaves 17,182 route-segment/time-period groups.

For the neighborhood comparison, we turn MUNI stop coordinates into points and create 400-meter buffers around them. Using GeoPandas and projected coordinates (EPSG:32610, UTM Zone 10N), we intersect those buffers with Census tract boundaries. We use the portion of each tract covered by a buffer to estimate nearby household and commuting characteristics. We then connect those estimates to segment summaries using the starting stop ID.

This assumes people are spread evenly within each tract, and a buffer can overlap several tracts. We should not describe it as an exact count of MUNI riders.

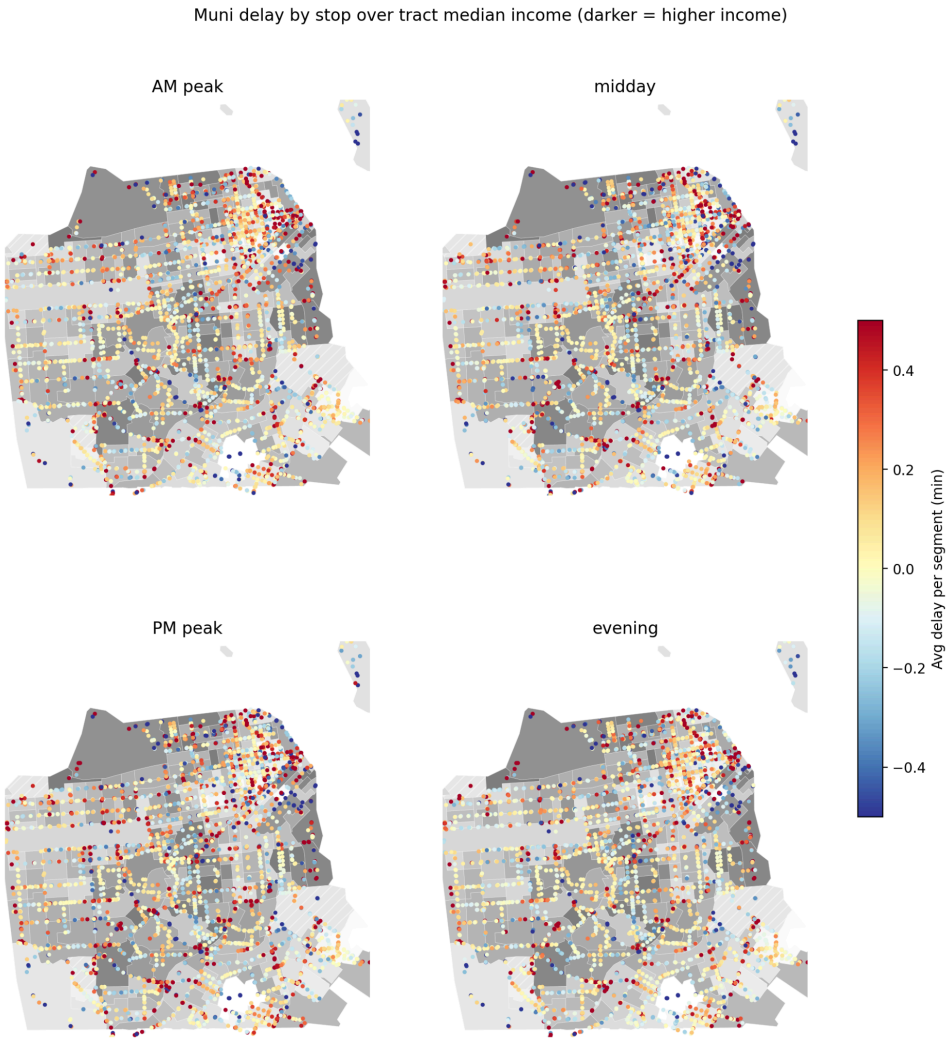
4. Early Results

We have already found meaningful differences between route segments and between times of day. Across the cleaned observations, the overall median travel-time difference is close to zero seconds, while the average is about 5.8 seconds longer than scheduled. That broad average does not mean every segment runs on time: some individual segments show much larger recurring delays.

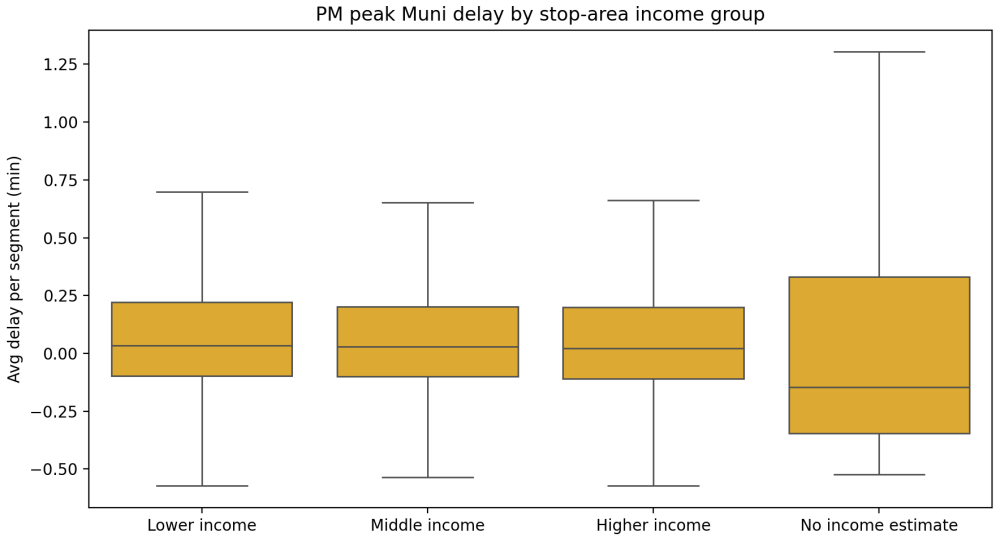
For example, in our current ranking, Route 54 from stop 16485 to stop 17606 has a median midday delay of about 19.7 minutes across 282 observations. The same segment also ranks high in the morning and afternoon periods. This is an early lead, not yet a final conclusion. We need to check whether that unusually long segment includes a layover, terminal operation, or data issue before calling it a routine travel delay.

Our code created neighborhood buffer estimates for all 3,234 mapped MUNI stops and joined them to route-segment summaries. We have not yet confirmed a relationship between delays and neighborhood income or transit commuting, so this part of the analysis is still exploratory.

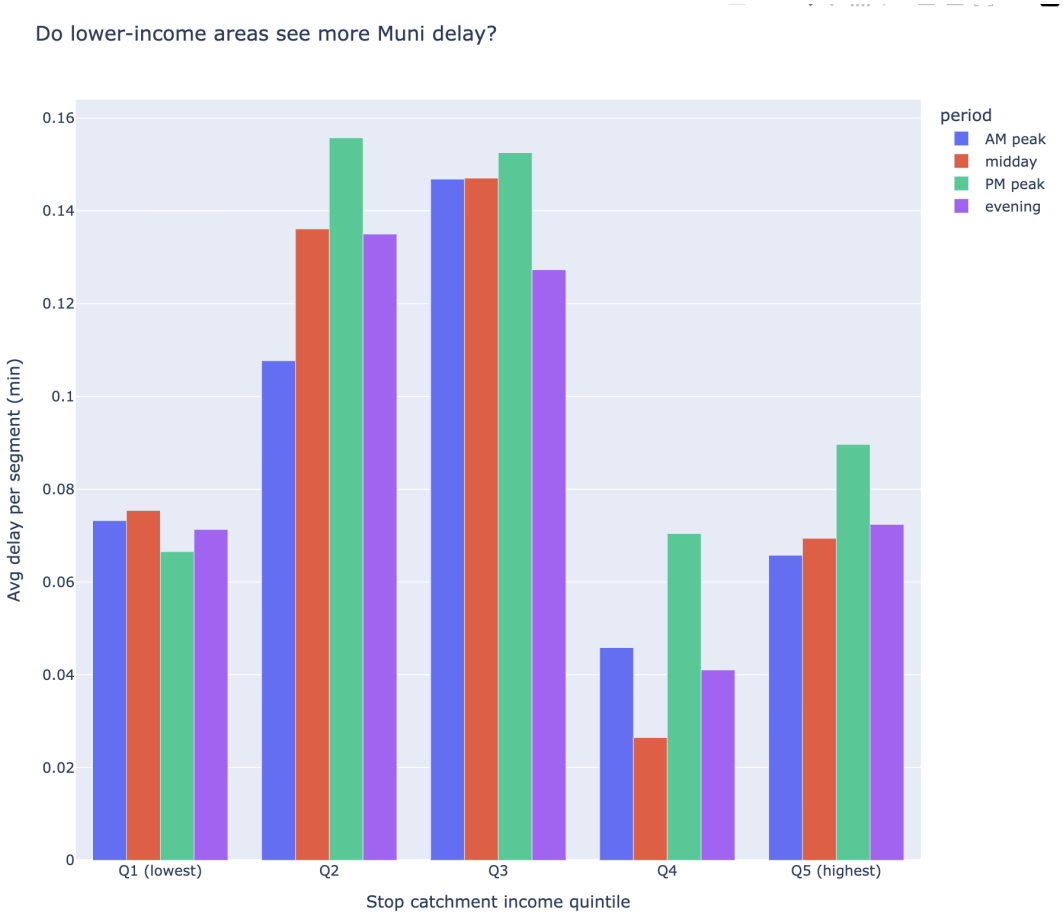
Figures to include or finalize



- Figure 1 — MUNI delay map: The map shows how MUNI travel-time delays vary across San Francisco during different times of day, with some stops experiencing longer delays than others.



- Figure 2 — PM peak delays by income group: The chart shows that lower-, middle-, and higher-income areas have similar median travel-time delays during the afternoon peak, although the amount of variation differs.



- Figure 3 — Delays by income quintile: The chart suggests that average MUNI delays are not consistently highest in the lowest-income areas, with some middle-income groups experiencing greater delays.

5. Plan to Finish and Questions for Feedback

The main data work is in place. Our next steps are to make sure the results are trustworthy and turn them into a clear story about where MUNI loses time.

Remaining task	Owner	Target date
Check highest-delay segments for layovers and data errors	Matthew Lim	Nov. 12, 2026
Finish and label at least 1 map and 3–5 charts	Johnluke Gonzales	Nov. 16, 2026
Test clean run, relative paths, and data download steps	Alex Chein	Nov. 20, 2026
Write figure captions and revise report interpretation	Sosun Yim	Nov. 25, 2026
Review final ZIP, GitHub documentation, and track needs	All team members	Dec. 2, 2026

Our biggest risks are very large or misleading delays, the size of the original transit data, and the uncertainty in using Census tract averages to represent people near individual stops. We can reduce these risks by checking suspicious segments against the source observations, documenting each filter, keeping all paths relative, and being careful not to claim that transit observations measure ridership.

We would like feedback on two questions: (1) Is our 400-meter stop-buffer approach a reasonable way to compare MUNI reliability with nearby neighborhood characteristics? (2) How should we handle high-delay segments that may include layovers or other nonstandard operations?